Swedish market

🇸🇪 Swedish market at a glance (2026)

Sweden is the "gold mine" for EV flexibility. The grid is stressed due to high wind energy volatility, and the government just opened the **Reduktionsplikt** (GHG mandate) to EV charging.

1. The "E-Credits" revenue (Regulatory)

This is "passive income" for CPOs. They get paid just for selling renewable electricity to cars.

- **Mechanism (Reduktionsplikt)** Fuel companies (like Preem, Circle K) must lower their CO2 impact. They buy "credits" from CPOs to offset their fossil fuel sales.
- **"4x Multiplier":** Under EU **RED III** rules, every 1 kWh of renewable electricity is counted as **4 kWh** toward the goal → *why it matters:* This effectively quadruples the value of the credit for the CPO.
- **The math:**

$$\text{Total Revenue} = \underbrace{(\text{Total kWh} \cdot \text{Hourly RES} - \text{E\%} \cdot 4)}_{\text{Compliance Volume}} \cdot \underbrace{(\text{Avoided CO2 per kWh} \cdot \text{CO2 Market Price})}_{\text{Market Price per e-Credit}}$$

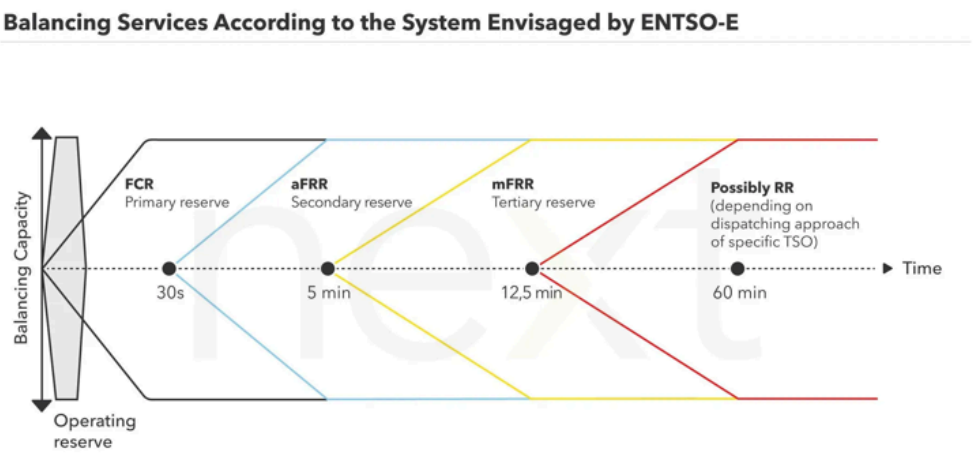
  - *Total kWh:* comes from raw charging data audited from CPO logs
  - *Hourly RES-E % (Renewable Energy Sources - Electricity)*: the renewable share of the grid during the charging event
  - *x4 multiplier:* "boost" provided by the EU Directive (RED III) to incentivize transport electrification
  - *Avoided CO2 per kWh:* extra margin Tether captures by auditing for the Carbon Intensity (CI). We use publicly available real-time models that determine the Grid CI per hour (change in emissions caused by an extra unit of demand in that hour). Then, the formula goes as follows:
    - Fossil Baseline CI: 94g CO2e / MJ (approx. 338.4g CO2e / kWh)
    - Verified Grid CI: sourced from database, eg. 15g CO2e / kWh
    - **Avoided CO2 per kWh:** Fossil Baseline CI - Verified Grid CI = 338.4-15 = 323.4g
  - *CO2 Market Price:* specific price per kilogram of CO_2 that an Obligated Party (fuel supplier) is willing to pay to avoid a regulatory penalty. Determined by the opportunity cost of fuel companies - €0.34 per kg of CO2 can be assumed as standard market price for 2026

Hourly RES-E% and Hourly Carbon Intensity values: Green Grid Compass

- it's already split between the 4 zones of Sweden's Energy Market

2. The "Flexibility" revenue (Grid Balancing)

This is "active income." Tether helps the CPO "pause" or "slow" charging for seconds/minutes to keep the grid stable.



The above is an illustration of the "order" of activation of balancing services (frequency reserves), in case of a grid event (ie. when the frequency in the power grid changes from its equilibrium state due to some unforeseen circumstances). Reserves differ mainly based on how fast they need to be activated and for how long does the activation need to last (eg. for FCR, you must automatically respond to grid signals within seconds and sustain that response for up to 20–30 minutes to help keep the power system balanced). From the CPO's perspective, the only key detail to know is that each reserve has its own market - therefore by operating in multiple ones, Tether can "stack" revenues.

The two markets that are interesting for us:

- **FCR (Frequency Containment Reserve)**
  - The grid needs to stay at exactly 50Hz. If it wobbles even slightly, it needs a "heartbeat" correction within seconds to *contain* it - this is what FCR is for.
  - For Tether, we tell the grid: *"We have 1,000 chargers. If you have a tiny emergency, we can instantly pause them for 30 seconds."* The grid pays us **just for being ready.** Even if we never actually pause the charging, we get paid an "availability fee" for every hour we are on standby. It's like an **Insurance Policy**. The grid pays us a premium just to know we are there if things go wrong.
  - *Status:* Prices are getting really low as the market saturates, but it remains a reliable baseline.
  - *Note:* FCR has several sub-markets, the ones we are interested in right now is **FCR-D Up** (ie. where we can turn charging *on* if needed) and **FCR-D Down** (ie. where we can *stop* the charging if needed). It is possible to simultaneously bid to both of these, so their potential revenues can be added up.
- **mFRR (manual Frequency Restoration Reserve)**
  - This is for bigger, longer grid problems that need to be *restored* (eg. a wind farm stops producing for 15 minutes). The grid needs a large "chunk" of power to be moved.
  - For Tether, we look at our AI predictions and say: *"We can shift 5 MegaWatts of charging from 6:00 PM to 6:15 PM without the drivers even noticing."* We get paid **a "standby fee" for being ready as well as a "delivery fee"** for the actual amount of energy we shifted. Because these events are rarer and more intense, the payouts can be very high.
  - *Note:* for the sake of the calculator, we only want to consider the "standby fee" (capacity remuneration) and not the "delivery fee" (activation remuneration), as activations depend on grid events and are generally very rare, whereas the standby fee is a recurrent revenue.
  - *Status:* In 2025, Sweden moved to 15-minute settlement periods. This created **huge price spikes** that EVs are perfect for catching.

FCR & mFRR price database: Mimer | Svenska kraftnät

- use FCR-D Up, FCR-D Down and mFRR CM